Design Guidelines for Thyristor Gate Drivers

Requirements, Trade-offs and Example Circuits for Phase-Control and Line-Commutated Applications

Abstract

This note collects the practical requirements that govern the design of a gate driver (gate unit) for phase-control and line-commutated thyristors, from small stud-mount devices up to large press-pack SCRs used in MW-class rectifiers and soft starters. The gate–cathode characteristic and the datasheet limits are reviewed first, followed by the requirements on pulse amplitude, slope, duration and off-state bias. Isolation, auxiliary supply, protection, layout and diagnostics are then treated. Four representative driver topologies are proposed and discussed, and a worked numerical example is given for a 6.5 kV/2.6 kA press-pack device. The document closes with a design checklist.

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1 Scope and Terminology

The subject of this note is the *gate unit*: the electronic circuit that converts a logic-level firing command, referenced to the control ground, into a gate current pulse referenced to the cathode of a thyristor floating at line potential.

Three functions are always present and should be designed separately:

(i) **Signal isolation** — transferring the firing command across the insulation barrier without distortion and with adequate common-mode transient immunity.

(ii) **Energy delivery** — sourcing the gate charge with the required amplitude, slope and duration.

(iii) **Protection and off-state control** — keeping the device firmly off when no command is present, and preventing spurious triggering.

The note is written for *phase-control thyristors* (PCT) and *fast/inverter-grade thyristors* that are turned off by circuit commutation. Gate-turn-off devices (GTO, IGCT) are outside the scope: they require a fundamentally different, low-inductance turn-off path and are covered by dedicated literature. Light-triggered thyristors (LTT) are mentioned only where relevant for the isolation discussion.

2 The Gate: Physical Behaviour and Datasheet Parameters

2.1 The gate–cathode junction

Electrically, the gate–cathode path is a large-area, heavily interdigitated p–n junction shunted by the distributed emitter short-circuit structure of the cathode. It therefore behaves as a diode in parallel with a resistance, with a strongly non-linear I_G–V_G characteristic and a wide unit-to-unit spread. It is *not* a capacitive load like a MOSFET or IGBT gate: the driver must deliver sustained current, not just charge.

Triggering occurs when the injected gate current raises the current gain of the *npn* section far enough to satisfy the regenerative condition $\alpha_{pnp} + \alpha_{npn} \geq 1$ in the region adjacent to the gate. Conduction then *spreads laterally* from that region at a plate velocity of the order of 50 µm/µs to 100 µm/µs. This spreading phenomenon is the physical reason why the gate pulse shape, and not only its amplitude, determines the tolerable anode $\mathrm{d}i/\mathrm{d}t$.

2.2 The trigger diagram

The datasheet parameters that define the required driver output characteristic are:

V_{GT}, I_{GT} — Gate trigger voltage and current: the values that guarantee firing, specified at 25 °C and often also at $T_{vj,\min}$.

V_{GD}, I_{GD} — Gate non-trigger voltage and current: the values below which the device is guaranteed *not* to fire, specified at $T_{vj,\max}$. The driver output in the off state must stay below these values with margin.

I_L, I_H Latching and holding current. The gate pulse must last until the anode current has exceeded I_L.

V_{FGM}, I_{FGM}
Peak forward gate voltage and current.

V_{RGM} Peak reverse gate voltage — typically only 5 V to 10 V. This is the most frequently violated limit in practice.

P_{GM}, $P_{G(AV)}$
Peak and average gate power dissipation, the latter usually specified over a 20 ms window.

Plotting these limits on the V_G–I_G plane produces the classical trigger diagram of Fig. 1. The design rule is simple to state:

> *The load line of the gate driver, drawn on the trigger diagram, must pass entirely through the "certain firing" area over the whole operating temperature range, and must never cross the P_{GM}, V_{FGM} or I_{FGM} boundaries.*

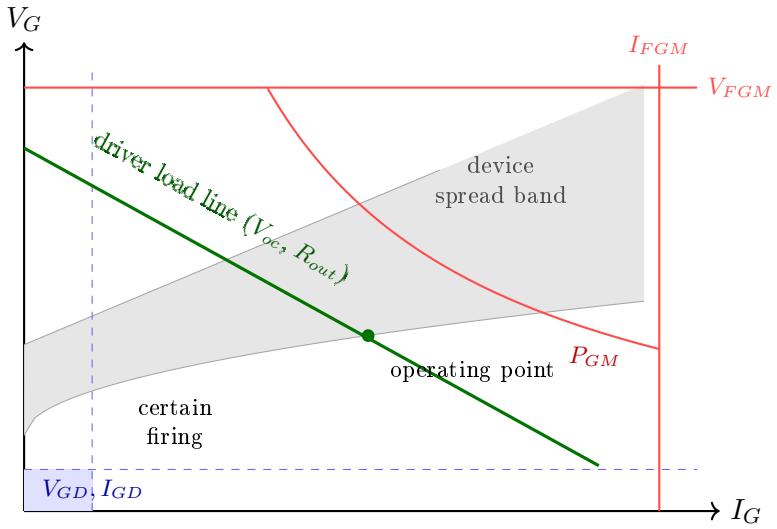


Figure 1: Thyristor gate trigger diagram. The driver load line must intersect the device spread band inside the certain-firing area and stay below the P_{GM}, V_{FGM} and I_{FGM} boundaries. The blue region is the guaranteed non-triggering area at $T_{vj,\max}$.

2.3 Temperature and spread

Two effects dominate the sizing:

- I_{GT} **increases at low temperature**, typically by a factor of 2 to 4 between 25 °C and −40 °C, because the emitter shorts divert a larger fraction of the gate current at low carrier lifetime. **The driver must be dimensioned at** $T_{vj,\min}$, i.e. for cold start-up, not at rated junction temperature.
- V_{GD} and I_{GD} **decrease at high temperature**. The off-state margin must therefore be verified at $T_{vj,\max}$.

Unit-to-unit spread on I_{GT} within one production type is commonly a factor of three or more. Designing for the typical value is a guaranteed field failure; the datasheet maximum at the minimum operating temperature is the correct starting point.

3 Requirements for the Gate Pulse

3.1 Amplitude: hard firing

The minimum requirement is $I_G > I_{GT}(T_{vj,\min})$, but this is never used in practice. The accepted design rule is

$$I_{G,\text{peak}} \;\geq\; 3 \ldots 10 \times I_{GT}\big(T_{vj,\min}\big), \tag{1}$$

with the upper end of the range used whenever high anode $\mathrm{d}i/\mathrm{d}t$ is expected. The reason is the lateral spreading mechanism: a large gate current turns on a larger initial cathode area, so the initial current density — and therefore the local dissipation during the first microseconds — is lower. The $\mathrm{d}i/\mathrm{d}t$ rating quoted on a datasheet is *always* conditional on a specified gate drive; using a weaker drive invalidates it.

For large press-pack devices, $I_{G,\text{peak}}$ in the range 2 A to 5 A is typical, against I_{GT} values of a few hundred milliamperes.

3.2 Slope

The gate current rise rate is as important as the amplitude:

$$\frac{\mathrm{d}i_G}{\mathrm{d}t} \;\geq\; 1\,\mathrm{A/\mu s} \quad \text{(typ. } 1\,\mathrm{A/\mu s} \text{ to } 5\,\mathrm{A/\mu s} \text{ for hard drive)}. \tag{2}$$

Equivalently, a rise time $t_r < 1\,\mu\mathrm{s}$, and preferably 0.1 µs to 0.5 µs, measured *at the device terminals*, not at the driver output. This is the specification that constrains the gate loop inductance:

$$L_{loop} \;\leq\; \frac{V_{drive} - V_{GT} - R_G I_G}{\mathrm{d}i_G/\mathrm{d}t}. \tag{3}$$

With a twisted pair of roughly 1 µH/m, this immediately limits the practical cable length between gate unit and device.

3.3 Duration, latching and back porch

The pulse must persist until the anode current has safely exceeded the latching current I_L. Three cases must be distinguished:

- **Resistive load, short pulse.** A pulse of 20 µs to 100 µs is adequate.
- **Inductive load.** The anode current rises slowly; the required pulse can be hundreds of microseconds to several milliseconds. A short pulse will cause intermittent misfiring, typically only at light load — a classic commissioning problem.
- **Discontinuous conduction, free-wheeling paths, series/parallel connection.** The safest approach is to keep the gate driven for the whole expected conduction interval.

The standard waveform is therefore a *hard pulse followed by a back porch* (Fig. 2): a short, high-amplitude spike that guarantees fast, wide-area turn-on, followed by a low plateau of $1 \ldots 2\times$ I_{GT} that holds the device fired until latching is certain, while keeping the average gate dissipation within $P_{G(AV)}$.

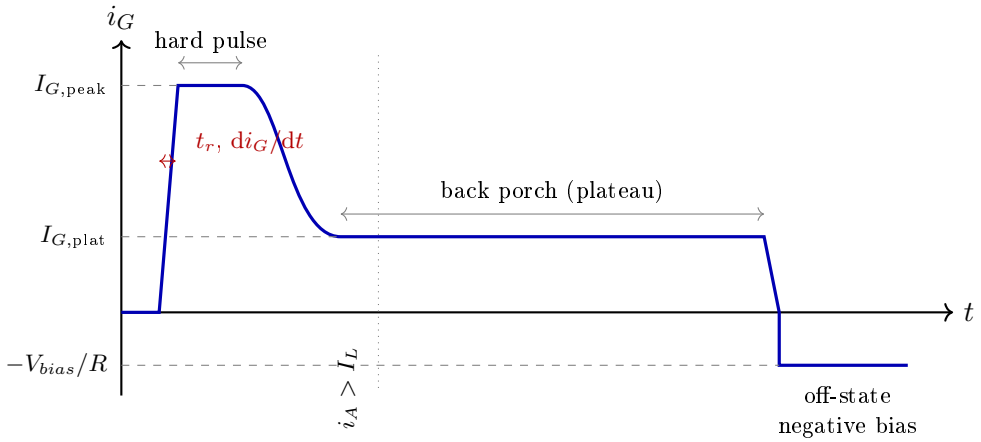


Figure 2: Recommended gate current waveform: a hard initial pulse for fast, large-area turn-on, a back porch that holds the device until latching, and a negative (or zero) bias in the off state.

3.4 Off-state bias and $\mathrm{d}v/\mathrm{d}t$ immunity

When the thyristor is off but forward biased, the displacement current through the blocking junction capacitance, $i = C_j\,\mathrm{d}v/\mathrm{d}t$, flows partly into the gate region and can trigger the device. Countermeasures, in order of effectiveness:

1. A **gate–cathode resistor** R_{GK} mounted *at the device*, typically $10\,\Omega$ to $100\,\Omega$ for large devices and up to $1\,\mathrm{k}\Omega$ for small ones. It diverts the displacement current and the junction leakage away from the gate. It also loads the driver, and this must be included in the sizing.

2. A **negative gate bias** of $-2\,\mathrm{V}$ to $-5\,\mathrm{V}$ between pulses. This is the most robust solution but requires a bipolar output stage and must respect V_{RGM} with margin, including all ringing.

3. A **low-impedance off state**: the output stage must present a low impedance to the gate when idle, not a high impedance.

Note that snubber design and $\mathrm{d}v/\mathrm{d}t$ immunity are coupled: a gate unit cannot compensate for an inadequate RC snubber.

3.5 Summary of target values

4 Isolation and Insulation Coordination

The gate unit straddles the main insulation barrier of the converter, and its insulation rating must be derived from the same coordination study as the rest of the equipment (IEC 60664-1, IEC 61800-5-1 for drives), considering:

- the **steady-state common-mode voltage** of the cathode with respect to control ground;
- the **repetitive peak** and the **impulse withstand** levels (BIL) of the installation;
- **creepage and clearance** for the pollution degree and altitude of the site;
- **partial discharge inception voltage** for medium-voltage equipment, which usually governs the design of the isolation transformer well before the dielectric withstand does;

Table 1: Typical design targets for a phase-control thyristor gate driver. Datasheet values always override these figures.

Quantity	**Typical target**	**Remark**
$I_{G,\mathrm{peak}}$	$3 \ldots 10 \times I_{GT}(T_{\min})$	Upper end for high $\mathrm{d}i/\mathrm{d}t$
$\mathrm{d}i_G/\mathrm{d}t$	1 A/µs to 5 A/µs	Measured at the device terminals
t_r	0.1 µs to 1 µs	Sets the gate loop inductance budget
Hard pulse width	10 µs to 50 µs	Covers turn-on delay and spreading
$I_{G,\mathrm{plat}}$	$1 \ldots 2 \times I_{GT}$	Limited by $P_{G(AV)}$
Plateau duration	until $i_A > I_L$	Whole conduction interval for inductive loads
Off-state bias	0 V to −5 V	Never exceed V_{RGM}, ringing included
R_{GK}	10 Ω to 100 Ω	Mounted at the device, not at the driver
Jitter	< 1 µs	Critical for series stacks and current sharing
Command-to-gate delay	matched within 1 µs	Across all channels of a bridge

- the **common-mode transient immunity** (CMTI), i.e. the $\mathrm{d}v/\mathrm{d}t$ that the barrier sees during commutation. The coupling capacitance across the barrier, C_{io}, injects $i = C_{io}\,\mathrm{d}v/\mathrm{d}t$ into the control electronics. Minimising C_{io} (electrostatic shield between windings, fibre optics) is the primary defence.

Three isolation strategies are used:

Table 2: Comparison of isolation strategies for the firing command.

Strategy	**Advantages**	**Limitations**
Pulse transformer	Carries command and energy together; no local supply needed; simple, robust, proven	Volt-second limit constrains pulse length; leakage inductance limits $\mathrm{d}i_G/\mathrm{d}t$; core reset required; C_{io} of the order of pF
Optocoupler + isolated supply	Arbitrary pulse length; easy shaping; low cost	Isolation typically limited to a few kV; CMTI is a design constraint; requires an isolated auxiliary supply
Fibre optic + local supply	Essentially unlimited isolation; negligible C_{io}; ideal for series stacks and MV	Requires a local energy source (auxiliary transformer or self-powering); more components; fibre handling and ageing

For series-connected stacks, fibre optics is effectively mandatory, both for the insulation level and because the transmission delay must be matched across all levels to within a microsecond to avoid unequal voltage sharing during turn-on.

5 Auxiliary Power for the Gate Unit

The gate unit consumes, per firing, an energy of the order

$$E_{fire} \approx V_G I_{G,\mathrm{peak}} t_{hard} + V_G I_{G,\mathrm{plat}} t_{plat}, \tag{4}$$

which for a large device is a few millijoules — small in absolute terms, but it must be available with a low source impedance to satisfy Eq. (2). Options:

1. **Isolated DC/DC converter** with a high-isolation transformer feeding the floating side. Straightforward, but the transformer becomes the insulation-critical component and its inter-winding capacitance directly degrades CMTI.
2. **Auxiliary winding on the main transformer** or a dedicated low-power potential transformer, common in large phase-control rectifiers.
3. **Self-powering from the anode–cathode voltage**, harvesting energy during the blocking interval through a high-voltage series resistor or, more efficiently, from the RC snubber current. This removes the barrier entirely for the power path, but requires careful start-up handling, since no energy is available before the converter is energised.
4. **Local storage capacitor** sized for several consecutive pulses, so that the pulse current is drawn from the capacitor and not from the supply. This is what actually determines the achievable $\mathrm{d}i_G/\mathrm{d}t$: place the capacitor immediately at the output stage.

6 Circuit Proposals

Four topologies are proposed, in increasing order of capability and cost.

6.1 Circuit A — Pulse transformer with peaking network

This is the reference solution for single devices and short-to-medium pulses (Fig. 3). A low-side MOSFET applies the auxiliary rail to the transformer primary. The DC blocking capacitor C_b provides automatic core reset and blocks any DC component caused by a stuck-on driver, which would otherwise saturate the core and destroy the MOSFET.

On the secondary, the peaking network shapes the two-level waveform of Fig. 2: at the leading edge C_p short-circuits R_p, so the gate current is limited only by R_s and the peak value is high; as C_p charges, the current settles to the plateau value set by $R_s + R_p$. Design equations:

$$I_{G,\mathrm{peak}} \approx \frac{nV_{cc} - V_{GT} - V_D}{R_s}, \qquad I_{G,\mathrm{plat}} \approx \frac{nV_{cc} - V_{GT} - V_D}{R_s + R_p} - \frac{V_{GT}}{R_{GK}}, \tag{5}$$

$$\tau_p = C_p \left(R_p \parallel R_s\right), \qquad t_{hard} \approx 3\tau_p, \tag{6}$$

where n is the turns ratio and V_D the drop of the series blocking diode D_s. The volt-second capability of the core sets the maximum pulse length:

$$A_e \;\geq\; \frac{V_{cc}\, t_{pulse}}{N_1 \Delta B}. \tag{7}$$

Notes.

- The series diode D_s prevents reverse gate current during core reset and protects the gate against the negative excursion; without it, V_{RGM} is easily exceeded.
- The leakage inductance of the transformer appears in series with the gate loop and directly limits $\mathrm{d}i_G/\mathrm{d}t$. Bifilar or interleaved winding is required; but interleaving increases C_{io}, so the two requirements must be traded off explicitly.

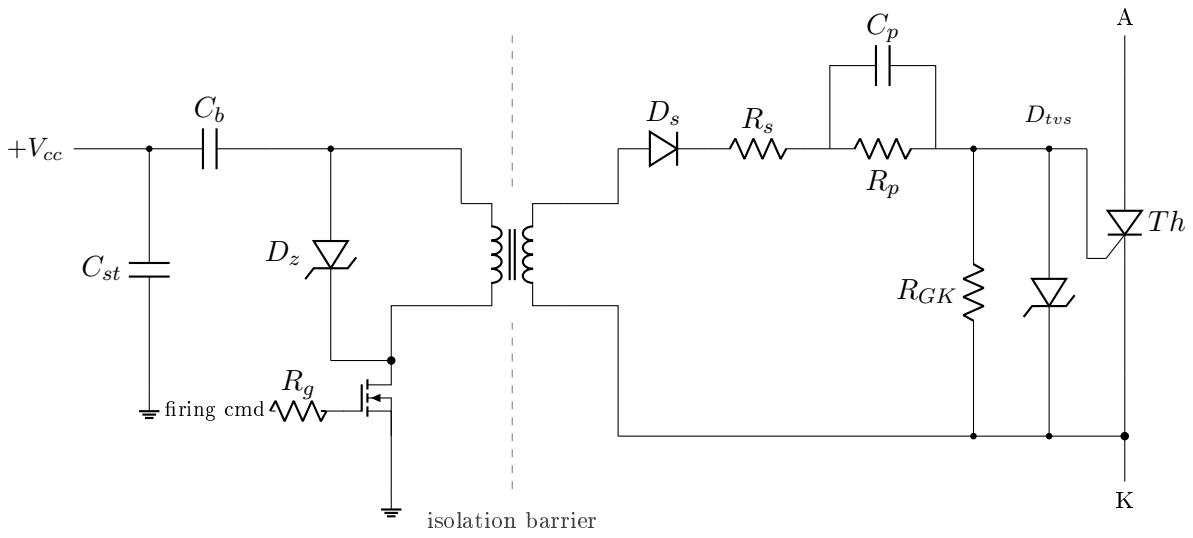


Figure 3: Circuit A — pulse transformer gate driver with DC blocking capacitor C_b for core reset, primary clamp D_z, and secondary peaking network $R_s/R_p/C_p$ producing the hard-pulse-plus-back-porch waveform. R_{GK} and the TVS are mounted at the device terminals.

- The local storage capacitor C_{st} must be placed at the primary of the transformer, with minimum loop area to the MOSFET.

6.2 Circuit B — Chopped (pulse-train) driver

Equation (7) makes long pulses expensive: a 500 µs pulse at 24 V requires a core roughly twenty times larger than a 25 µs pulse. The standard solution is to *chop* the pulse at 5 kHz to 20 kHz (Fig. 4). The transformer then only sees the volt-seconds of one carrier period, while the envelope can be arbitrarily long.

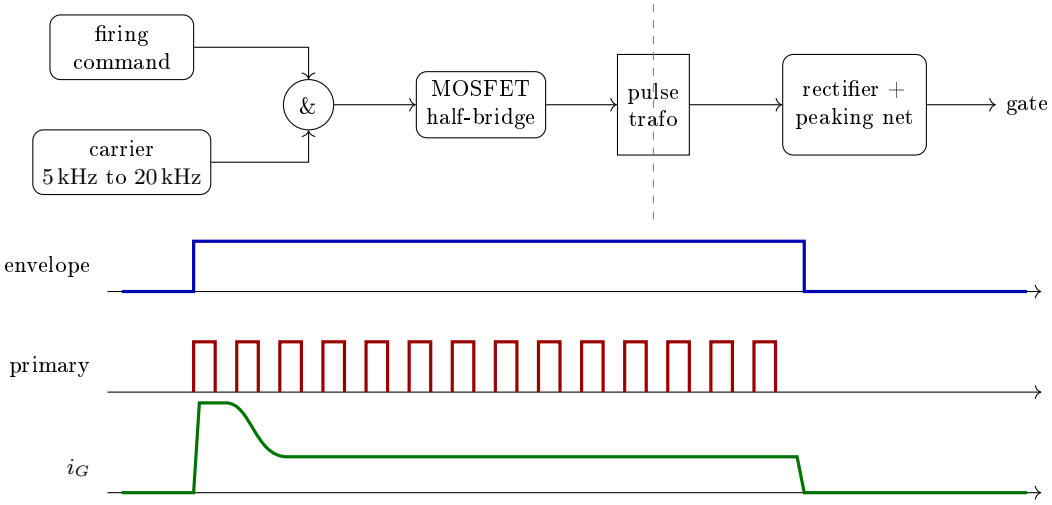


Figure 4: Circuit B — chopped gate driver. The firing envelope is modulated by a 5 kHz to 20 kHz carrier so that the transformer only handles the volt-seconds of a single carrier period. Light filtering on the secondary reconstructs a quasi-continuous gate current.

The secondary filtering must be a compromise: enough capacitance to smooth the plateau between carrier pulses, little enough not to slow the leading edge. A common arrangement is a fast rectifier feeding the peaking network of Circuit A, with a small filter capacitor placed

after the peaking network. A half-bridge (or full-bridge) primary is preferred over a single-ended stage, since it doubles the usable flux swing and provides intrinsic core reset.

6.3 Circuit C — Fibre-optic gate unit with two-level current source

For medium-voltage equipment and series stacks, the command is transmitted over plastic optical fibre and the energy is stored locally (Fig. 5). The output stage is a two-level source: M_1 delivers the hard pulse from the storage capacitor C_{st} through the low resistance R_1; a monostable then releases M_1 and M_2 maintains the plateau through R_2 for as long as the optical command is present. Between pulses M_3 applies the negative bias and holds the gate at low impedance.

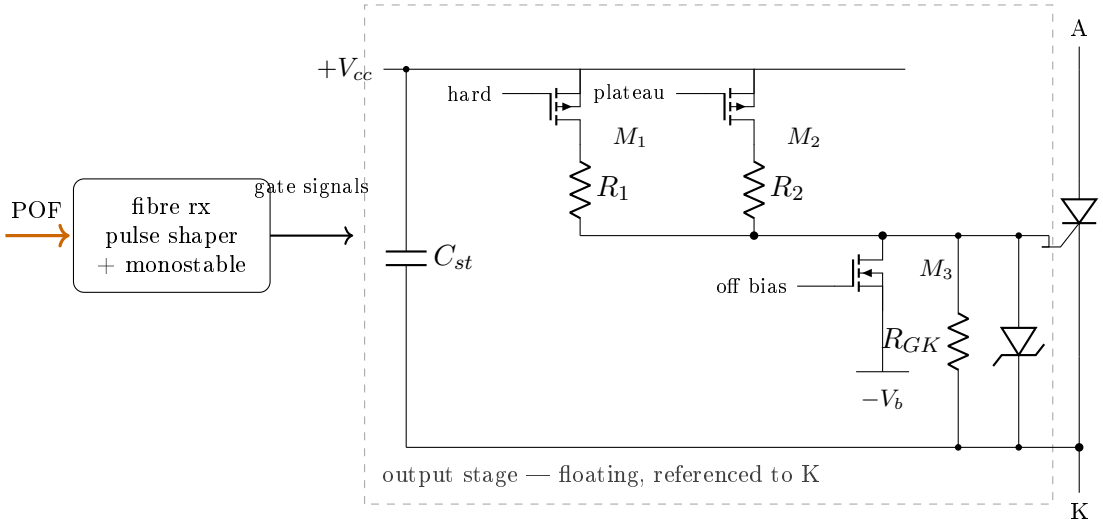


Figure 5: Circuit C — fibre-optic gate unit with local storage and a two-level output stage. M_1 sources the hard pulse through R_1, M_2 the back porch through R_2, and M_3 applies the negative off-state bias.

Notes.

- C_{st} must be a low-ESL type placed within a few millimetres of M_1; it, and not the auxiliary supply, determines the achievable $\mathrm{d}i_G/\mathrm{d}t$.
- The monostable that terminates the hard pulse should be retriggerable and should have a defined maximum on-time, so that a stuck-high optical input cannot exceed $P_{G(AV)}$.
- A watchdog on the optical link is essential: loss of fibre must be interpreted as “no fire”, never as “fire”. Use a modulated optical carrier rather than a static light level so that a broken fibre and a permanent-off command are distinguishable.

6.4 Circuit D — Self-powered gate unit

Where no auxiliary supply is available on the floating side — typically in high-voltage series stacks — the gate unit harvests its energy from the anode–cathode voltage (Fig. 6). A high-voltage current-limiting network in parallel with the snubber charges the local storage capacitor during the blocking interval; a shunt regulator clamps the storage voltage. The fibre carries only the command.

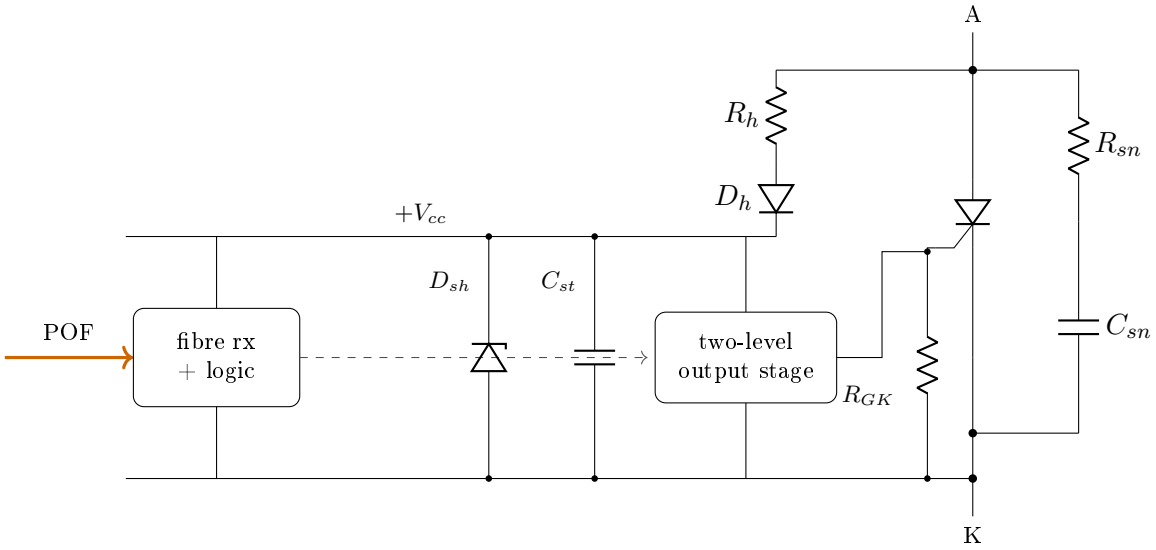


Figure 6: Circuit D — self-powered gate unit. Energy is harvested from the anode–cathode blocking voltage through R_h/D_h and stored in C_{st}, clamped by the shunt regulator D_{sh}. No auxiliary supply crosses the insulation barrier.

Notes.

- Start-up must be handled explicitly: the first firing after energisation can only occur once C_{st} is charged. Either the control must wait for a "ready" feedback, or the first cycles must be fired in a mode that does not require full gate energy.
- R_h dissipates continuously in the blocking state and must be rated for the full repetitive peak voltage. Harvesting from the snubber current instead of a resistive divider is considerably more efficient.
- In a series stack, the harvesting network is part of the static and dynamic voltage-sharing design and must be included in the sharing calculation.

7 Protection, Layout and EMC

7.1 Component-level protection

- R_{GK} **at the device.** Mandatory. Placed on the device terminals, not on the driver board.
- **Reverse-voltage protection.** A series diode or an anti-parallel diode limits the negative gate excursion. V_{RGM} of a few volts is easily exceeded by ringing in a long gate cable.
- **Bidirectional TVS** across gate and cathode, at the device, for transients coupled into the gate loop. Its clamping voltage must sit between the drive voltage and V_{FGM}/V_{RGM}.
- **Series gate resistance.** Beyond setting the current level, it damps the resonance between the gate loop inductance and the device gate capacitance and prevents oscillatory retriggering.
- **Output stage current limit or fuse** on the driver side, so that a short-circuited gate lead does not destroy the gate unit.

7.2 Gate loop layout

- Use a **twisted pair** for the gate–cathode connection; a shielded twisted pair, with the shield bonded at the driver end only, if the environment is harsh. Both conductors must be routed together over the entire length — a common error is to take the cathode return from a power terminal.
- Keep the **loop area** minimal. The gate loop is a receiving antenna for the $\mathrm{d}i/\mathrm{d}t$ of the main circuit.
- Keep the **cable length** within the budget of Eq. (3). Below 1 m is preferable; beyond a few metres the peaking network must be redesigned or the drive voltage raised.
- **Route the gate cable away from the main busbars** and cross them at right angles where crossing cannot be avoided.
- In a bridge, keep the **propagation delay matched** across all channels; a mismatch translates directly into unequal voltage sharing in series connections and unequal current sharing in parallel connections.

7.3 Common-mode considerations

The barrier capacitance C_{io} and the commutation $\mathrm{d}v/\mathrm{d}t$ define a common-mode current that returns through the control ground. Design measures: electrostatic shield in the isolation transformer connected to the control ground; common-mode chokes on the primary leads; fibre optics where the $\mathrm{d}v/\mathrm{d}t$ is severe. Verify by direct measurement with a common-mode transient generator rather than by calculation.

8 Diagnostics and Feedback

In industrial equipment the gate unit is usually expected to provide at least:

- **Pulse confirmation** — a feedback signal indicating that the gate current pulse actually flowed, obtained from a small current transformer or a shunt in the gate loop. This distinguishes a driver fault from a device fault.
- **Anode voltage monitoring** — a divider indicating whether the device is blocking or conducting. Used for turn-on verification, for detecting a shorted device before energisation, and for missing-pulse detection.
- **Supply undervoltage lockout** — inhibits firing when the local supply is below the level needed to meet the drive specification. Firing with insufficient gate current is more damaging than not firing at all, because turn-on occurs over a small area.
- **Overtemperature and watchdog** on the gate unit itself.

For series stacks, the pulse confirmation signals of all levels should be combined into a single "stack healthy" indication, since a single missing pulse imposes the whole stack voltage on one device.

9 Worked Example

Consider a large press-pack phase-control thyristor, roughly 6.5 kV/2.6 kA, in a six-pulse line-commutated bridge with an inductive load. Representative datasheet values (to be replaced by the actual ones):

I_{GT} at 25 °C	250 mA	max
I_{GT} at −40 °C	600 mA	estimated, ≈ 2.5×
V_{GT}	3 V	max
V_{GD} / I_{GD} at $T_{vj,\max}$	0.25 V / 10 mA	
I_L	2 A	
V_{FGM} / I_{FGM} / V_{RGM}	10 V / 8 A / 5 V	
P_{GM} / $P_{G(AV)}$	100 W / 15 W	

Step 1 — Pulse amplitude. From Eq. (1) with $I_{GT}(T_{\min}) = 0.6\,\mathrm{A}$ and a factor of 5: $I_{G,\mathrm{peak}} = 3\,\mathrm{A}$, comfortably below I_{FGM}. Peak gate power $\approx 5\,\mathrm{V} \times 3\,\mathrm{A} = 15\,\mathrm{W}$, well below P_{GM}.

Step 2 — Plateau. $I_{G,\mathrm{plat}} = 1.5 \times I_{GT}(25\,^\circ\mathrm{C}) \approx 0.9\,\mathrm{A}$. With a conduction interval of 6.7 ms at 50 Hz, the duty over a 20 ms window is 1/3, hence $P_{G(AV)} \approx 3.5\,\mathrm{V} \times 0.9\,\mathrm{A} \times 1/3 \approx 1.1\,\mathrm{W}$, acceptable.

Step 3 — Peaking network. Take $V_{drive} = 15\,\mathrm{V}$ and $V_{GT}+V_D = 3.5\,\mathrm{V}$, so 11.5 V is available. From Eq. (5):

$$R_s = \frac{11.5\,\mathrm{V}}{3\,\mathrm{A}} \approx 4\,\Omega, \qquad R_s + R_p = \frac{11.5\,\mathrm{V}}{0.9\,\mathrm{A} + 0.11\,\mathrm{A}} \approx 11.4\,\Omega \Rightarrow R_p \approx 7.5\,\Omega,$$

where 0.11 A is the current diverted by $R_{GK} = 27\,\Omega$ at 3 V. For a hard-pulse duration of 20 µs:

$$\tau_p = \frac{t_{hard}}{3} \approx 6.7\,\mu\mathrm{s}, \qquad C_p = \frac{\tau_p}{R_p \parallel R_s} = \frac{6.7\,\mu\mathrm{s}}{2.6\,\Omega} \approx 2.6\,\mu\mathrm{F} \;\rightarrow\; 2.2\,\mu\mathrm{F}.$$

Step 4 — Gate loop inductance. Requiring $\mathrm{d}i_G/\mathrm{d}t = 3\,\mathrm{A/\mu s}$ and using Eq. (3) with the full 11.5 V available at $t = 0^+$:

$$L_{loop} \le \frac{11.5\,\mathrm{V}}{3\,\mathrm{A/\mu s}} \approx 3.8\,\mu\mathrm{H}.$$

At 1 µH/m for a twisted pair, and reserving 1.5 µH for the transformer leakage and the layout, the cable is limited to roughly 2 m. Longer runs require either a higher drive voltage or a local gate unit.

Step 5 — Transformer core. For a directly transmitted pulse of 500 µs at 24 V with $N_1 = 20$ and $\Delta B = 0.3\,\mathrm{T}$, Eq. (7) gives $A_e \ge 2000\,\mathrm{mm}^2$ — impractical. Chopping at 20 kHz with 50 % duty reduces the requirement to $A_e \ge 100\,\mathrm{mm}^2$, an ordinary small core. This single calculation is the reason Circuit B is the standard choice whenever long pulses are needed.

10 Design Checklist

1. Datasheet limits collected: I_{GT}, V_{GT} at $T_{vj,\min}$; V_{GD}, I_{GD} at $T_{vj,\max}$; I_L, I_H; V_{FGM}, I_{FGM}, V_{RGM}; P_{GM}, $P_{G(AV)}$.
2. Driver load line plotted on the trigger diagram and verified at both temperature extremes.
3. $I_{G,\mathrm{peak}}$ and $\mathrm{d}i_G/\mathrm{d}t$ consistent with the conditions under which the $\mathrm{d}i/\mathrm{d}t$ rating is specified.
4. Pulse duration verified against the worst case: minimum load current, maximum load inductance, minimum supply voltage.
5. $P_{G(AV)}$ verified at maximum firing frequency and maximum conduction angle.
6. Off-state condition verified: $V_{GK} < V_{GD}$ at $T_{vj,\max}$, with R_{GK} fitted.

7. V_{RGM} verified including ringing, measured at the device.
8. Insulation: creepage, clearance, impulse withstand, partial discharge, C_{io} and CMTI.
9. Transformer volt-second capability verified at maximum pulse width, maximum supply voltage and maximum temperature; core reset mechanism present.
10. Gate loop inductance budget verified against cable length.
11. Undervoltage lockout, watchdog and fail-safe interpretation of a lost command.
12. Channel-to-channel delay and jitter measured, not assumed.
13. Verification performed on the actual device at the coldest specified temperature, with the weakest expected supply, and with the gate current measured *at the device terminals*.

Notes on Verification

Two measurement practices are worth stating explicitly:

- Measure the gate current with a current probe placed on the gate lead at the *device*, and the gate voltage differentially at the *device* terminals. What the driver output does is of secondary interest; the loop inductance and the cable are part of the circuit.
- Test at the extremes, not at nominal. A gate driver that works on the bench at 25 °C with a new device and a short lead tells very little about behaviour at −25 °C, with a worst-case device and the installed cable length. Most field misfiring problems are traced to one of these three margins.

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